Proof. First remove the definable locus where the output RV-tuple is not locally constant in the input VF-coordinates. On each resulting open fiber with both input and output RV-tuples fixed, extend the VF-map by zero outside its domain and apply [17, Corollary 9.9] to its partial derivatives. Their nonexistence and discontinuity loci are definable and have VF-dimension less than n in every such fiber. Retaining or projecting away RV-coordinates does not increase VF-dimension, so the combined exceptional locus still has VF-dimension less than n. Off this locus all partial derivatives, and hence their determinant, are continuous. In arity zero the Jacobian is the empty determinant 1. □

Resolved referee R035 [Gap] – Equal VF dimensions and generic RV constancy.
Original source line 993. State $n = n'$ in Lemma 2.38; otherwise the determinant called the Jacobian has not been defined. Moreover, the preceding existence of one full-dimensional fiber for each input t does not show that such fibers cover almost every input point. Use the standard generic local constancy of a definable map from VF^n to $\mathrm{RV}^{m'}$ first, and then apply the cited differentiability theorem on its open fibers. This provides the missing almost-everywhere coverage.

Author resolution. The author approves the square-dimension hypothesis $n = n'$ and the generic local-constancy argument for the output RV-labels. The proof now applies the cited partial-differentiability theorem on open fixed-label charts and explains the smaller-dimensional exceptional locus. The arity-zero Jacobian is the empty determinant 1.

Terminology 2.39. We say that a set $I \subseteq \Gamma_\infty^n$ is *γ-bounded*, where $\gamma \in \Gamma$, if it is contained in the box $[\gamma, \infty]^n$, and is *doubly γ-bounded* if it is contained in the box $[-\gamma, \gamma]^n$. More generally, let A be a subset of $\mathrm{VF}^n \times \mathrm{RV}_\infty^m \times \Gamma_\infty^l$ and

$$\mathrm{valim}(A) = \{(\mathrm{val}(a), \mathrm{vrv}(t), \gamma) \mid (a, t, \gamma) \in A\} \subseteq \Gamma_\infty^{n+m+l},$$

where $\mathrm{val}(a)$ and $\mathrm{vrv}(t)$ are taken coordinatewise. We say that A is *γ-bounded* if $\mathrm{valim}(A)$ is, and likewise for *doubly γ-bounded*. A set is *bounded* if it is γ-bounded for some $\gamma \in \Gamma$, and *doubly bounded* if it is doubly γ-bounded for some $\gamma \in \Gamma$. The notation A_Γ continues to mean projection to existing Γ-coordinates.

Resolved referee R036 [Clarification] – A valuation image is not a coordinate projection.
Original source line 1001. The notation A_Γ here conflicts with Notation 2.1, where it denotes projection to existing Γ-coordinates. Use a separate notation such as $\mathrm{valim}(A)$ for this coordinatewise valuation image. Also explicitly say that 'bounded' means γ-bounded for some γ, and similarly for 'doubly bounded'.

Author resolution. The author approves $\mathrm{valim}(A)$ for the coordinatewise valuation image, reserving A_Γ for projection to existing Γ-coordinates. The existential finite bound in "bounded" and "doubly bounded" is now explicit. This coordinatewise convention is distinct from the minimum tuple valuation used in differentiation.

Definition 2.40 (Contractions)**.** A function $f : A \longrightarrow B$ is rv-*contractible* if there is a (necessarily unique) function $f_\downarrow : \mathrm{rv}(A) \longrightarrow \mathrm{rv}(B)$, called the rv-*contraction* of f, such that

$$(\mathrm{rv} \upharpoonright B) \circ f = f_\downarrow \circ (\mathrm{rv} \upharpoonright A).$$

Similarly, it is res-*contractible* (respectively, val-*contractible*) if the same holds in terms of res (respectively, val or vrv, depending on the coordinates) instead of rv.

Lemma 2.41. *Let $U \subseteq \mathrm{RV}^k$ be a doubly bounded set and $f : U \longrightarrow \mathrm{RV}$ a definable function. Then $f(U)$ is also doubly bounded. The same holds if the codomain of f is Γ.*

Proof. For an RV-valued f, put $h = \mathrm{vrv} \circ f$; for a Γ-valued f, put $h = f$. We retain the induction on k, with the case $k = 0$ immediate. For $k > 1$ and $t \in \mathrm{pr}_1(U)$, the inductive hypothesis makes $h(U_t)$ doubly bounded. Its finite bound

$$\gamma_t = \sup\{|h(t, x)| \mid x \in U_t\}$$

exists by definable completeness of Γ, and the least-upper-bound formula makes $t \longmapsto \gamma_t$ definable. The projection $\mathrm{pr}_1(U)$ is doubly bounded, so the unary Γ-valued case bounds all γ_t and hence bounds $h(U)$. Thus both claims reduce to the case $k = 1$ and a definable function $g : U \longrightarrow \Gamma$. Then, by C-minimality, we may assume that g vrv-contracts to a function $g_\downarrow : \mathrm{vrv}(U) \longrightarrow \Gamma$. Since definable functions in Γ are piecewise $\mathbb{Q}$-linear, the range of $g_\downarrow$ must be doubly bounded. □

Resolved referee R037 [Gap] – Uniformity in the bounded RV-image lemma.
Original source line 1017. The induction on k must control the bounds uniformly in the remaining coordinates; boundedness on each fiber alone does not give a bound on their union. A precise route is to consider the relation of possible values $\mathrm{vrv}(f(u))$ over $\mathrm{vrv}(u) \in \Gamma^k$. Residue-field/value-group orthogonality makes each such fiber finite, uniformly; stable embeddedness and elimination in divisible ordered abelian groups then give finitely many piecewise rational-affine branches. These have bounded images on a bounded box. Include this justification, or an equivalent explicit uniform induction.

Author resolution. The author retains the original induction. Its fiber bounds are chosen definably as finite suprema of the absolute output values; the unary Γ-valued case then bounds this definable bound function uniformly. The proof now spells out that step, without replacing the induction by a relation of all input and output values.

Definition 2.42. Let $A \subseteq \mathrm{VF}^n$ and $f : A \longrightarrow \mathcal{P}(\mathrm{VF}^m)$ be a definable function whose range is bounded. For $a \in \mathrm{VF}^n$ and $L \subseteq \mathrm{VF}^m$, we say that L is a *limit set of f at a*, written as $\lim_{A \to a} f \subseteq L$, if for every $\epsilon \in \Gamma$ there is a $\delta \in \Gamma$ such that if $c \in \mathfrak{o}(a, \delta) \cap (A \smallsetminus a)$ then $f(c) \subseteq \bigcup_{b \in L'} \mathfrak{o}(b, \epsilon)$ for some $L' \subseteq L$.

A limit set L of f at a is *minimal* if no proper subset of L is a limit set of f at a.

Referee R038 [Clarification; revised] – Simplify the limit-set definition.
Original source line 1046. Say explicitly that $\bigcup_{a \in A} f(a)$ is bounded. The auxiliary subset L' does no work: the displayed condition is equivalent to $f(c) \subseteq \bigcup_{b \in L} \mathfrak{o}(b, \epsilon)$. The following definition of continuity is applied to finitary functions without a global boundedness hypothesis, so the present definition needs a compatible extension. One can use the same formula for arbitrary correspondences whenever a finite affine limit exists, or formulate limits in $(\mathbb{P}^1(\mathrm{VF}))^m$ from the outset; local boundedness near the point suffices for affine local uses.

Observe that if $\lim_{A \to a} f \subseteq L$ and $b \in L$ is not isolated in L then actually $\lim_{A \to a} f \subseteq L \smallsetminus b$. So in a minimal limit set every element is isolated. Moreover, if a minimal limit set L exists then its topological closure is unique:

Lemma 2.43 ([17, Lemma 9.2])**.** *Let $L_1, L_2 \subseteq \mathrm{VF}^m$ be two minimal limit sets of f at a. Then the closures of L_1, L_2 coincide.*

This lemma justifies writing $\lim_{A \to a} f = L$ when L is a closed (hence the unique) minimal limit set of f at a, in particular, when L is finite and minimal.

Let $A \subseteq \mathrm{VF}^n$ be a definable set with $\dim_{\mathrm{VF}}(A) = n$ and $f : A \longrightarrow \mathcal{P}(\mathrm{VF}^m)$ a definable finitary function. We say that f is *continuous* if $\lim_{A \to a} f = f(a)$ for all $a \in A$.

Lemma 2.44. *The function f is continuous almost everywhere.*